Ali Hamza (Corresponding Author)
Department of Software Engineering, SEECS
National University of Sciences and Technology (NUST)
Islamabad, Pakistan
Email: ahamza.msse25mcs@student.nust.edu.pk
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Subject: Manuscript Submission — "Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations"

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript titled **"Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations"** for consideration as a regular research article in *Expert Systems with Applications* (ESWA).

Operational flood decision-support systems must fuse heterogeneous multi-modal data streams (satellite radar, optical imagery, reanalysis climate forecasts, terrain rasters, gauge networks, and unstructured bulletins) under frequent sensor dropout and inter-source disagreement. Contemporary LLM-arbitrated agentic pipelines suffer from four critical operational vulnerabilities: (1) minority-class collapse on critical inundation states, (2) uncalibrated probability estimation and silent hallucination under missing modalities, (3) lack of per-agent credibility propagation, and (4) absence of source-level provenance trails.

In this work, we present **AEGIS** (Agentic Evidential Geographic Intelligence for Sustainability), a novel framework interfacing Subjective Logic operator algebra with a specialist five-agent catalog over a shared four-state Dirichlet frame. The key contributions of our study include:

- **Evidential Multi-Agent Orchestration**: Combining Consensus & Compromise Fusion (CCF) and Weighted Belief Fusion (WBF) driven by a dynamic Jensen–Shannon divergence signal.
- **Monotonic Uncertainty Guarantee**: Utilizing a partial-observable projection rule that mathematically guarantees epistemic uncertainty u grows monotonically as modalities drop.
- **Brier-Score Credibility Propagation**: Online tracking of agent reputation to weight fusion decisions dynamically without manual threshold tuning.
- **Provenance-Aware Explainability**: A DuckDB-backed provenance ledger powering an interactive spatial dashboard for source-level auditing.

We evaluate AEGIS on the benchmark Brisbane February–March 2022 flood fold (200 cells $\times$ 24 days $=$ $4,800$ cell-day instances). AEGIS-SL achieves an F1-Macro of 0.7190, AUROC of 0.9310, and Expected Calibration Error (ECE) of 0.043, outperforming monolithic raw-feature late-fusion baselines and demonstrating strong robustness against minority-class collapse where LLM arbiters fail.

We confirm that:

1. This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere.

2. All co-authors (Ali Hamza and Ghulam Mujtaba) have read, approved, and agreed to the submission of this manuscript to *Expert Systems with Applications*.

3. There are no financial or personal conflicts of interest to declare.

Thank you for considering our work. We look forward to hearing from you regarding the review process.

Sincerely,

Ali Hamza
Corresponding Author
National University of Sciences and Technology (NUST), Islamabad, Pakistan